

CYSE 301: Cybersecurity Technique and Operations

Assignment 2: Sword vs. Shield

In this assignment, you will act as an attacker to identify the vulnerabilities in the LAN network and a defender to apply proper countermeasures. You need to provide a screenshot for each task below.

Each student needs to login into the **CCIA virtual environment** using the link <https://cyber.lab.cova-cci.org/> to complete this assignment. You can find the VM required for this lab under **Hyper-V Manager** after logging to CCIA.

AI Use Policy for This Assignment

You are required to use an AI assistant (e.g., Claude, ChatGPT) at several points in this assignment. Each AI task is worth its own points.

Rules of engagement:

- AI may help you interpret scan output, explain firewall rule logic, and draft written evaluations.
- AI may NOT run Nmap, configure pfSense, or perform any technical step for you — you must generate your own scan output, rules, and screenshots on your own VMs.
- You must critically evaluate every AI response against your own results, not simply agree with it. Screenshot the AI conversation for each question.
- An AI-generated evaluation that just restates the AI's answer without your own critical judgment will not receive full credit.

Task A: Sword - Network Scanning (5 + 15+ 10 + 10 = 40 points)

Power on/ Connect the following listed VMs,

- External Kali
- pfSense
- Ubuntu
- Windows Server 2022
- Internal Kali

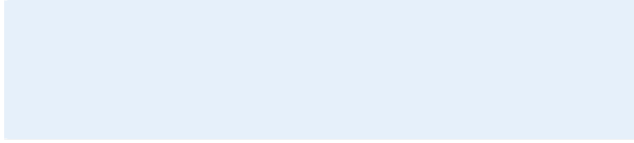
Make sure not to add/delete any firewall rule/policies before continuing.

Screenshot required — label it with the exact step/question number.

1. Run Wireshark in the Internal Kali VM while External Kali is scanning the network.

2. Use **Nmap in External Kali** to profile the basic information about the **subnet** topology (including open ports information, operating systems, etc.)

[Add the screenshot here]



You need to submit a table summarizing the following (please add more rows if required):

IP address Of the VMs	Open ports	Service versions	Operating System detection	Backend software information

3. **(AI Task) (10 points):** Copy your Nmap output into an AI system (ChatGPT or equivalent).

Please write a **200-word critical evaluation** answering the following, comparing the actual result with that got from prompting AI:

- What was the exact prompt you used
- What was AI's full response
- Did AI correctly interpret open ports?
- Did it assume vulnerabilities without evidence?
- Did it exaggerate risk?
- What did it miss?
- Would you trust this analysis in a real SOC environment?

4. **Using Chain-of-Thought Prioritization. (10 pts)**

Using the Nmap output, provide multiple prompts asking the AI assistant for

- each open port
- reasoning about which ones are internet-facing risks vs. internal-only risks
- rank them from highest to lowest priority

Screenshot the full reasoning trace with your prompts (not just the final ranked list).

In **3–5 sentences**, state whether the AI's step-by-step reasoning matches your own understanding of the subnet and identify one step in the chain where the AI's logic was weak, unsupported, or skipped a step.

NOTE: To get full credit, you must submit the screenshot of your prompts given to AI tool. You should evaluate the AI response, not simply agree with it.

-----End of Task A-----

Task B: Shield – Protect your network with a firewall (10 + 15+ 15 + 20 = 60 points)

In order to receive full credits, you need to fill the table (add more rows if needed), implement the firewall rule(s), attach the screenshot of your rule configured in the firewall table, and attach the screenshot of the rule and the Ping test verification of the rules.

Important: Open Pfsense using a browser on the Internal Kali.

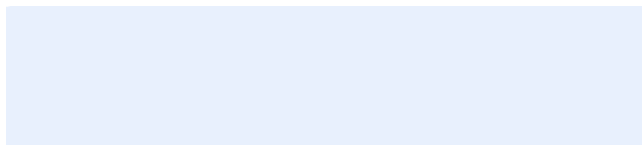
1. Configure the pfSense firewall rule to block the ICMP traffic from External Kali to the Ubuntu VM.

Rule #	Interface	Action	Source IP	Destination IP	Protocol (port # if applicable)

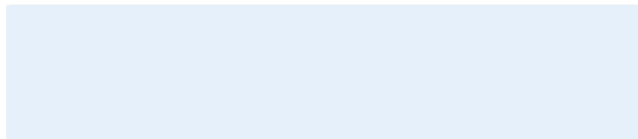
Attach Two Screenshots (Mandatory)-

- a. A screenshot of the firewall rule that you have created in Pfsense for the above task
- b. A screenshot of the terminal where you have tested the rule is in effect or not.

[Add the screenshot here]



[Add the screenshot here]



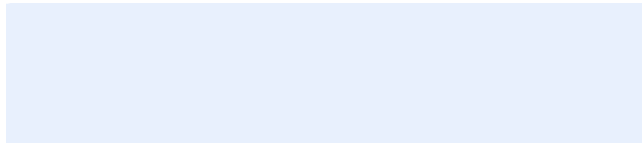
2. Clear the previous firewall policies and configure the pfSense firewall to block all ICMP traffic from External Kali to the LAN side.

Rule #	Interface	Action	Source IP	Destination IP	Protocol (port # if applicable)

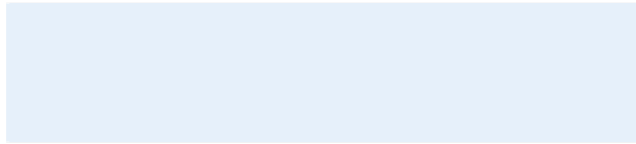
Attach Two Screenshots (Mandatory)-

- c. A screenshot of the firewall rule that you have created in Pfsense for the above task
- d. A screenshot of the terminal where you have tested the rule is in effect or not.

[Add the screenshots for Firewall Rules and Ping test here]



[Add the screenshot here]



Add your Reflection (~150 words):

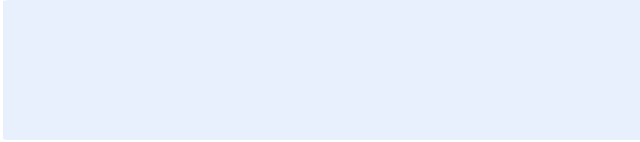
- What security advantages does blocking ICMP provide?
 - What operational disadvantages might occur?
3. Clear the previous firewall policies and configure the pfSense firewall to block ALL traffic from External Kali to the LAN side, except for the FTP protocol directed towards Ubuntu. You can add more rows in the table here, if necessary.

Rule #	Interface	Action	Source IP	Destination IP	Protocol (port # if applicable)

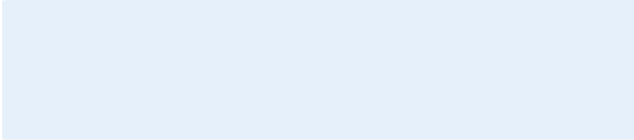
Attach Two Screenshots (Mandatory)-

- e. A screenshot of the firewall rule that you have created in Pfsense for the above task
- f. A screenshot of the terminal where you have tested the rule is in effect or not.

[Add the screenshot here]



[Add the screenshot here]



4. Keep the firewall policies you created in Task B.3 and repeat Task A.1 to rescan the subnet using nmap in External Kali.

Now, compare the results/ findings, and complete the following table and add your reflection as asked here:

Before Firewall Rules	After Firewall Rule
Open Ports	Eg. Filtered/Blocked
OS Detection Accuracy	
Service Enumeration	

Write in your own words (~ 200 words) explaining:

- a. Why does Nmap show “filtered” instead of “closed”, if there is any.
- b. How firewall behavior affects scan results
- c. What information is hidden from attackers

Extra credit (15 points):

Use NESSUS to enumerate the security vulnerabilities of the Microsoft Windows Server 2022 VM in the CCIA network and submit the following in a PDF file:

- Prioritize vulnerabilities
- Recommend remediation strategy
- Identify the highest real-world risk
- CVSS scores
- Your own judgment

-----End of Extra credit-----

Deliverables:

1. Screenshots for every numbered task/step in Tasks A and B, each labeled with its task/question number.
2. Screenshots of each AI conversation, labeled to match its question numbers.
3. Completed rule tables for Task B.1–B.3 and the before/after comparison table for B.4.
4. Written reflections for tasks A-3, A-4, B-2 B-4.
5. Extra credit Nessus report submitted as a separate PDF, if attempted.